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Dual-Gaussian Schottky barrier distribution in CVD graphene/n-Si diodes from temperature-dependent transport

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ABSTRACT

The transformative potential of graphene/Si Schottky junctions for electronic and power-electronic applications stems from their exceptional thermal and chemical resilience compared to conventional metal contacts. In this work, chemical vapor deposition-grown graphene was transferred onto n-type Si to fabricate Schottky barrier diodes, and the structural, morphological, and electronic properties of the graphene were characterized by Raman spectroscopy and ultraviolet photoelectron spectroscopy. The temperature-dependent current–voltage (I–V) characteristics of the fabricated devices were systematically investigated over the temperature range of 80–300 K. At room temperature, the ideality factor and Schottky barrier height (SBH) were determined to be 2.84 and 0.51 eV, respectively, based on the thermionic emission theory. An ideality factor significantly exceeding unity indicates that charge transport is not governed exclusively by thermionic emission, but also involves thermionic field emission and generation–recombination processes. The pronounced temperature dependence of both the ideality factor, decreasing from 7.41 ± 0.19 at 80 K to 2.84 ± 0.17 at 300 K, and the SBH, increasing from 0.17 ± 0.001 eV to 0.51 ± 0.005 eV over the same range, reveals the presence of significant Schottky barrier inhomogeneity at the graphene/silicon interface. The spatial non-uniformity of the barrier height was quantitatively analyzed within the potential fluctuation formalism of Werner and Güttler, assuming a Gaussian distribution of barrier heights. Two distinct temperature regimes were identified: region I (80–140 K) and region II (160–300 K), yielding mean barrier heights $\phi_m = 0.42$ and 0.72 eV, standard deviations $\sigma_s = 0.10$ and 0.05 eV, and Richardson constants $A^* = 21.2$ and 13.8 A cm^{-2} K^{-2}. The experimentally extracted Richardson coefficients are substantially below the theoretical value for bulk n-type Si (112 A cm^{-2} K^{-2}), attributed to the finite density of states arising from the linear band dispersion of graphene. The interface trap density, estimated from the excess ideality factor, lies in the range of ~ 2.6–9.0×10^{13} eV^{-1} cm^{-2} between 300 and 80 K, consistent with amphoteric Pb^0-type defects at the Si/SiO_2 interface. The series resistance remains nearly constant over most of the investigated temperature range, with a slight reduction observed at higher temperatures. All key electrical parameters were reproduced within approximately 5% across three nominally identical devices, confirming fabrication reliability and interface stability.

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INTRODUCTION

Graphene, a one-atom-thick crystalline sheet of carbon, represents the ultimate two-dimensional material. Its unique sp^2-hybridized lattice hosts delocalized π-electrons originating from the out-of-plane p_z orbitals, giving rise to exceptional electronic transport properties. In contrast to conventional metal–semiconductor contacts, graphene exhibits remarkable thermal robustness, chemical inertness, and environmental stability.[1] These attributes, combined with its tunable work function and atomically sharp interface with silicon, make graphene an ideal contact material for next-generation Schottky barrier diodes, enabling enhanced interface control, reduced Fermi-level pinning,

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and improved device reliability. Furthermore, the intrinsically high thermal conductivity of graphene, reaching up to ~5300 W m^{-1} K^{-1} for freely suspended, pristine graphene, though considerably reduced in CVD-grown samples due to grain boundary scattering and phonon coupling to the substrate, facilitates efficient heat dissipation, motivating the design of graphene-based Schottky barrier diodes for high-performance, high-temperature applications.[2] The rectifying characteristics and transport mechanisms of graphene/semiconductor Schottky junctions have been extensively investigated, highlighting their potential for low-power and high-frequency electronic applications.[3,4] In particular, graphene/Si Schottky junctions have been employed in high-performance photodetectors, including self-powered and broadband devices, owing to their efficient photocarrier separation and fast photoresponse.[5,6] Furthermore, graphene-based Schottky junctions can serve as functional elements in vertical hot-electron transistor architectures for high-frequency operations.[7] These application prospects highlight the importance of understanding the temperature-dependent charge transport and Schottky barrier inhomogeneity in graphene/Si junctions.

When brought into contact with a moderately doped n-type semiconductor, graphene, owing to its semi-metallic band structure, establishes a Schottky junction at the interface that governs interfacial charge transport. Graphene does not form direct covalent or ionic bonds with the substrate; instead, it couples to the semiconductor surface primarily through van der Waals interactions and charge-transfer processes induced by interface defects.[8] Charge transfer at the graphene/semiconductor junction occurs as a consequence of the work function mismatch, leading to Fermi-level alignment under thermal equilibrium.[9] Since the work function of graphene (~4.4–4.8 eV)[1] typically exceeds that of moderately doped n-type silicon (~4.2–4.3 eV),[1] electrons are transferred from the semiconductor into graphene upon contact, resulting in carrier depletion near the semiconductor surface. This charge redistribution induces upward band bending in the semiconductor and leads to the formation of a Schottky barrier at the graphene/n-type silicon interface.[9] The barrier height for an ideal graphene/Si junction is governed by the Schottky–Mott relation, though experimentally extracted values frequently deviate from this idea due to Fermi-level pinning, interface states, and barrier non-uniformity effects that have been described in detail in comprehensive treatments of Schottky barrier physics.

Despite the promise of graphene as a contact material, the graphene/silicon interface is intrinsically non-ideal. Residual metallic impurities, particularly copper originating from the Cu foil used in CVD graphene synthesis, can persist after the transfer process and contribute to interfacial disorder.[10] Silicon surfaces in graphene/Si Schottky diodes are also prone to a high density of dangling bonds, giving rise to amphoteric Pb^0-type interface states capable of trapping both electrons and holes. Even in the presence of an ultrathin interfacial oxide layer, charge trapping and de-trapping at these states remain active and significantly influence the transport characteristics.[8] These Pb^0 centers, stabilized at the Si/SiO_2 interface, facilitate trap-assisted tunneling and partial Fermi-level pinning, thereby degrading rectification characteristics despite the nominal passivation provided by the oxide. Owing to their amphoteric nature, these defects capture electrons under reverse bias and emit holes under forward bias, leading to dynamically evolving barrier inhomogeneities across the junction.[8] In addition, structural ripples and electronically active grain boundaries intrinsic to CVD-grown polycrystalline graphene introduce nanoscale electrostatic potential fluctuations, while the presence of the ultrathin interfacial oxide layer inhibits hydrogen passivation of the silicon surface, introducing additional interface dipoles that locally modify the graphene work function and enhance Schottky barrier height non-uniformity.[8] All these effects collectively lead to ideality factors greater than unity and a spatially inhomogeneous barrier landscape.[11] Understanding the temperature-dependent transport behavior arising from this inhomogeneity is, therefore, essential for the reliable integration of graphene/Si Schottky diodes into a broad range of electronic applications.

Previous studies have established the rectifying behavior, fundamental transport mechanisms, and temperature-dependent barrier inhomogeneity of graphene/n-Si Schottky junctions.[1,10,12–14] In particular, thermionic-emission, Landauer-type, and Werner–Güttler approaches have been employed to describe charge transport and barrier non-uniformity in these junctions.[1,10,12,13] However, a comprehensive correlation of the temperature-dependent ideality factor, Schottky barrier height, series resistance, interface-trap density, and Richardson constant with distinct transport regimes remains limited.

In this study, we report strongly rectifying Schottky interfaces formed by transferring large-area CVD-grown graphene onto moderately doped n-type silicon substrates and systematically investigate their temperature-dependent electrical characteristics over 80–300 K. Particular emphasis is placed on correlating the extracted electrical parameters with the observed two-regime barrier-distribution behavior, providing further insight into the temperature-dependent charge transport at the graphene/n-Si interface. Figure 1 depicts the energy-band profile of the graphene/n-Si junction formed upon contact.

EXPERIMENTAL DETAILS

The n-type silicon substrates (thickness ≈ 350 μm, donor concentration $N_D \sim 1 \times 10^{16}$ cm^{-3}) were cleaned using the standard RCA procedure to remove organic and ionic contaminants from the surface. Native oxide removal was subsequently carried out by immersing the substrates in dilute hydrofluoric acid (HF), followed by controlled re-oxidation in ambient atmosphere to intentionally form an ultrathin native SiO_2 interfacial layer for Schottky barrier modulation, a strategy well established for graphene/Si junction engineering. Ohmic back contacts were fabricated by depositing aluminum (Al) onto the rear surface of the Si substrates in an ultrahigh vacuum (UHV) chamber through a shadow mask. The ohmic linearity of the back contact was verified from the symmetry of low-voltage I–V measurements at all temperatures.

Monolayer graphene was synthesized on 25 μm thick Cu foil (Alfa Aesar, Item No. 13382) using an atmospheric pressure chemical vapor deposition (APCVD) technique in a split-tube furnace. Before growth, the Cu foil was annealed at 1020 °C for 30 min under an Ar (500 SCCM) and H_2 (30 SCCM) atmosphere. Subsequently, graphene growth was carried out at the same temperature by introducing CH_4 (3 SCCM) for 30 min. After growth,

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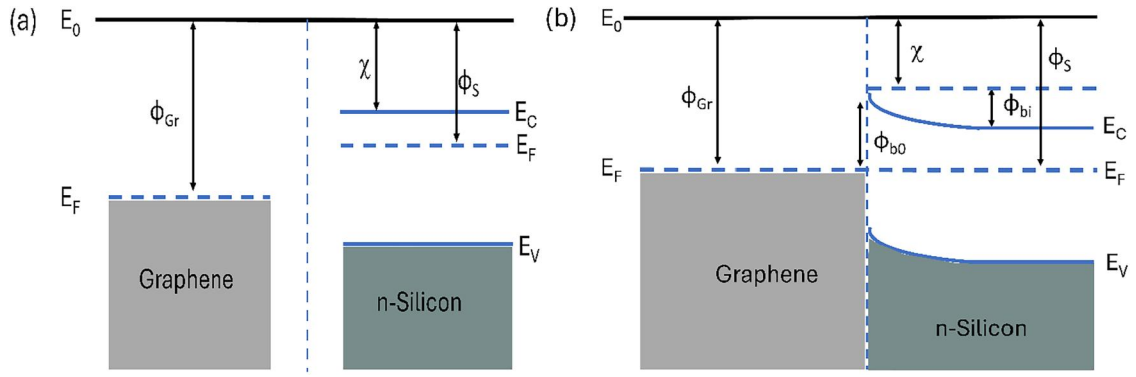


FIG. 1. Energy band profile of graphene and Si (a) before interface formation, (b) after interface formation with ϕ_{Gr}, ϕ_{Si}, χ, ϕ_{bi}, ϕ_{b0} representing the work function of graphene, the work function of silicon, the electron affinity of silicon, built-in potential, and the Schottky barrier height (SBH), respectively.

the CH_4 flow was stopped, and the furnace was rapidly cooled to room temperature by opening the furnace while maintaining the Ar and H_2 flow.[15] The graphene was subsequently transferred onto the prepared n-Si substrates using a standard polymer-assisted wet transfer technique.[16] The resulting graphene/Si Schottky junction defined an active device area of $1 \times 1\ mm^2$. Electrical contacts to the graphene layer were formed by using silver paste as glue for copper wires onto the graphene region supported by SiO_2, ensuring electrical isolation from the Schottky junction area and avoiding parasitic leakage paths, following a contact geometry analogous to that described by Luongo *et al.*[17]

The structural quality and vibrational properties of the synthesized graphene films were investigated using Raman spectroscopy (inVia Reflex, Renishaw, UK). The electronic properties of graphene were further investigated by ultraviolet photoelectron spectroscopy (UPS) using a Thermo Scientific ESCALAB QXi system (Thermo Fisher Scientific, USA) with He I excitation source ($h\nu = 21.22$ eV).

Temperature-dependent I–V measurements were performed using an Agilent B1500A semiconductor parameter analyzer over the temperature range of 80–300 K in steps of 20 K, under vacuum in a cryogenic probe station. To assess device-to-device reproducibility, three nominally identical Gr/Si Schottky diodes were fabricated and characterized under identical conditions. The extracted room-temperature electrical parameters, including the ideality factor, Schottky barrier height, series resistance, and rectification ratio, agreed within approximately 5% across all three devices, confirming good fabrication uniformity and the statistical reliability of the temperature-dependent trends reported in this work.

RESULTS AND DISCUSSION

The Raman spectrum of graphene transferred onto the SiO_2/Si substrate is presented in Fig. 2(a). The prominent G ($\sim 1585\ cm^{-1}$) and 2D ($\sim 2686\ cm^{-1}$) bands, along with a high (I_{2D}/I_G) ratio of ~3.6 and a narrow symmetric 2D band (FWHM $\approx 30\ cm^{-1}$), confirm the formation of high-quality monolayer graphene. Furthermore, the weak D band ($\sim 1350\ cm^{-1}$) with an (I_D/I_G) ratio of ~0.4 indicates a low defect density, with the residual defects likely originating from grain boundaries and the transfer process.[18] To further investigate the electronic properties of the graphene layer, UPS measurements were performed. As shown in Fig. 2(b), the secondary electron cutoff (SECO) was determined to be 16.20 eV, yielding a work function of 5.02 eV using He-I excitation ($h\nu = 21.22$ eV). It is worth noting that CVD-grown graphene synthesized on Cu foil generally exhibits p-type conductivity due to charge transfer from adsorbed oxygen/moisture species and residual transfer-induced doping.[19] The measured work function of 5.02 eV, which is higher than that of pristine graphene (~4.5 eV), corroborates the p-type nature of the transferred graphene and confirms its suitability as a transparent Schottky electrode for the fabricated graphene/Si heterojunction device.[19]

The transferred CVD-grown monolayer graphene exhibits p-type character and, owing to its high carrier concentration, behaves as a p^+-like contact. When interfaced with n-type Si, the depletion region is primarily confined to the silicon, and the resulting graphene/Si junction follows the transport characteristics of a Schottky contact. Therefore, the current–voltage characteristics can be described using the conventional thermionic emission model.

Figures 3(a) and 3(b) show the device schematic employed for two-terminal I–V measurement and the room-temperature forward I–V characteristics of the Gr/Si Schottky barrier diode, respectively. The experimental I–V data were fitted using the standard thermionic emission (TE) model, which can also be derived within the Landauer transport formalism,[20]

$$I = AA^*T^2 \exp\left(\frac{-q\phi_{b0}}{k_B T}\right)\left[\exp\left(\frac{qV}{\eta k_B T}\right) - 1\right], \quad (1)$$

where q is the electronic charge, A is the diode area ($1\ mm^2$), and A^* is the effective Richardson coefficient (theoretical value for n-type Si: $112\ A\ cm^{-2}\ K^{-2}$).[1,12] k_B is the Boltzmann constant, η is the ideality factor, and ϕ_{b0} is the effective zero-bias Schottky barrier height. Fitting Eq. (1) to the room-temperature data yields

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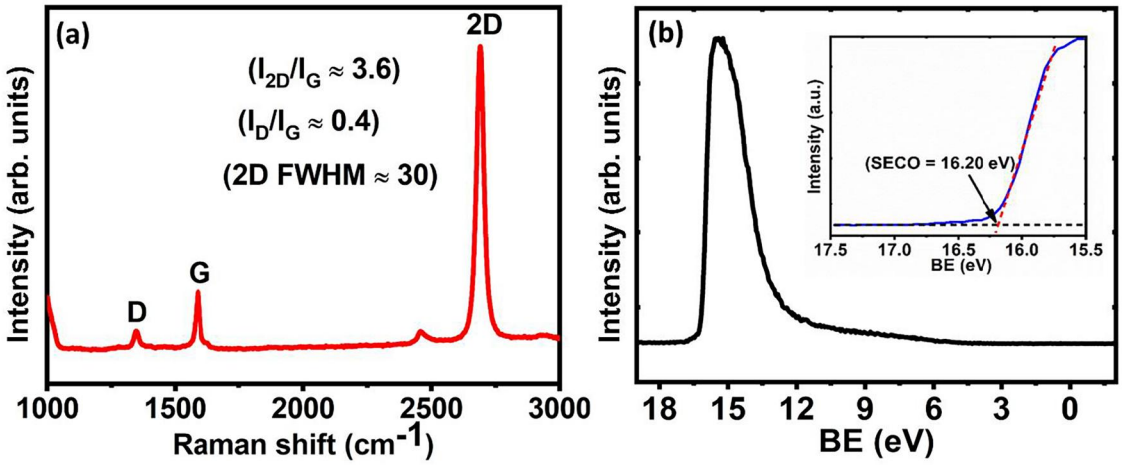


FIG. 2. (a) Raman spectrum of the transferred CVD-grown graphene exhibiting prominent D, G, and 2D bands. The high I_{2D}/I_G ratio (~3.6) together with the narrow 2D-band FWHM (≈30 cm^{-1}) confirms the formation of high-quality monolayer graphene. (b) UPS spectrum of graphene. The inset shows the secondary electron cutoff (SECO) region used to determine the graphene work function (ϕ_{Gr}) of approximately 5.02 eV.

an ideality factor of $\eta = 2.84$ and a Schottky barrier height of $\phi_{b0} = 0.51$ eV, which is in good agreement with the previously reported value of ~0.49 eV.[7]

To account for the voltage drop across the diode series resistance at high forward bias, Eq. (1) was extended to a TE + R_s form, $I = AA^*T^2\exp(-q\phi_{b0}/k_BT)[\exp(q(V - IR_s)/\eta k_BT) - 1]$, and the room-temperature forward I–V characteristics [the same dataset shown in Fig. 3(b)] were re-fitted over the full measured bias range using this expression. The fit yields a series resistance $R_s \approx 1.65\,k\Omega$, in good agreement with the value obtained independently from the Cheung and Cheung method (Fig. 5). This agreement indicates that the departure from ideal thermionic-emission behavior at higher forward bias is dominated by the ohmic voltage drop across R_s rather than by an additional transport mechanism. For the temperature-dependent analysis presented below, η and ϕ_{b0} were extracted from the quasi-linear ln(I)–V region unaffected by series-resistance roll-off, while $R_s(T)$ was evaluated separately using the Cheung and Cheung method, as described below.

According to the Schottky–Mott model,[21] the barrier height for an ideal graphene/semiconductor contact is given by

$$\phi_b = \phi_{Gr} - \chi_{Si}, \tag{2}$$

where ϕ_{Gr} is the work function of graphene and χ_{Si} is the electron affinity of silicon (~4.05 eV).[1] Due to unintentional doping and environmental sensitivity, the work function of CVD graphene

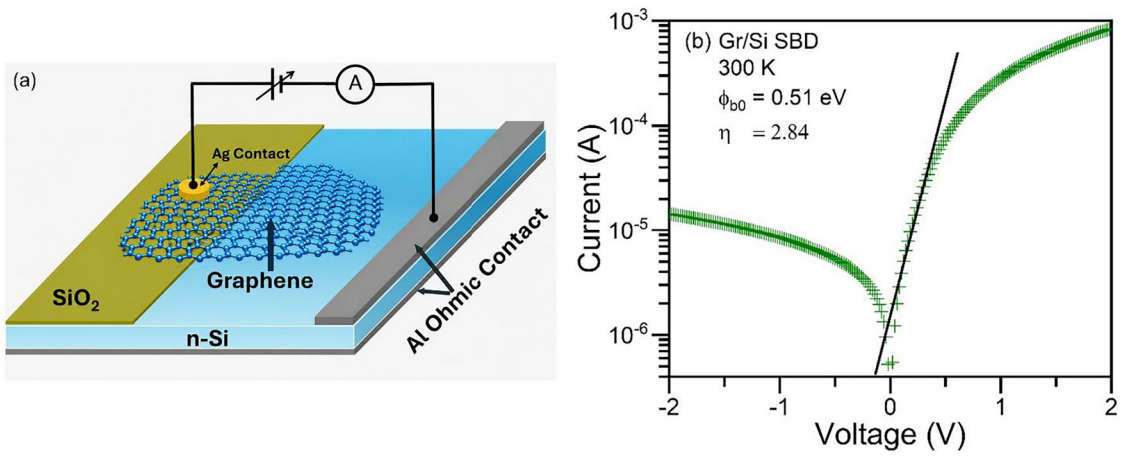


FIG. 3. (a) Schematic of the Gr/Si Schottky barrier diode employed for two-terminal I–V measurements. (b) Room-temperature I–V characteristics of the Gr/Si Schottky barrier diode fitted using the thermionic emission model [Eq. (1)], yielding an ideality factor $\eta = 2.84$ and a Schottky barrier height $\phi_{b0} = 0.51$ eV.

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varies between approximately 4.4 and 4.8 eV,[1] giving an expected ideal barrier height in the range of 0.35–0.75 eV. The experimentally extracted SBH of 0.51 eV falls within this range, confirming the formation of a well-defined rectifying Schottky contact. The room-temperature rectification ratio, defined as $R = I_{forward}/I_{reverse}$ at $|V| = 1$ V, is approximately 10^2, consistent with values reported for CVD graphene/Si diodes by Luongo *et al.*[17] The transport behavior of graphene/semiconductor Schottky junctions under such non-ideal conditions has been comprehensively reviewed by Di Bartolomeo,[3] and the underlying Schottky barrier physics treated in depth by Tung.[22] The observed ideality factor $\eta = 2.84$ ($\eta > 1$) indicates that charge transport at room temperature is not governed exclusively by thermionic emission, additional mechanisms including thermionic field emission, direct tunneling, and generation-recombination processes contribute to the measured current.[22]

To characterize the temperature dependence of transport, I–V measurements were carried out from 80 to 300 K in steps of 20 K. Figures 4(a) and 4(b) show the temperature-dependent I–V characteristics and the extracted ideality factor and zero-bias SBH at each temperature, respectively. The leakage current at 80 K is of the order of $\sim 10^{-7}$ A, increasing to $\sim 10^{-5}$ A at 300 K, consistent with thermally activated transport across the junction. The rectification ratio decreases with increasing temperature, primarily due to enhanced thermally generated reverse-bias leakage current.[17,23,24] In graphene-based junctions, this effect is further amplified by temperature-induced Fermi-level shifts in graphene under reverse bias, which reduce the effective barrier height, and by image-force barrier lowering at the moderate doping level of the Si substrate.

The ideality factor decreases from 7.41 ± 0.19 at 80 K to 2.84 ± 0.17 at 300 K, while the apparent SBH increases from 0.17 ± 0.001 eV to 0.51 ± 0.005 eV over the same temperature range. This inverse correlation between η and ϕ_{b0} is a well-established signature of Schottky barrier inhomogeneity and has been widely reported for both conventional metal/semiconductor and graphene/semiconductor contacts.[1,8,10] At lower temperatures, transport is dominated by current flowing through localized regions with lower barrier heights, whereas at higher temperatures, a greater fraction of the junction area participates in conduction, raising the apparent mean barrier height and reducing the effective ideality factor.[1]

The characteristic tunneling energy E_{00} for the n-Si substrate was estimated using the standard expression,[25]

$$E_{00} = \frac{\hbar}{2}\sqrt{\frac{qN_D}{m^*\varepsilon_s}}, \tag{3}$$

where $N_D \sim 1 \times 10^{16}$ cm^{-3}, effective electron mass $m^* \approx 0.26\ m_0$, and $\epsilon_s \approx 11.7\ \epsilon_0$[26,27] for silicon. This yields $E_{00} \approx 2.7$ meV. At 300 K, $E_{00}/k_BT \approx 0.10 \ll 1$, confirming that thermionic emission dominates transport at room temperature. At 80 K, $k_BT \approx 6.9$ meV and $E_{00}/k_BT \approx 0.39$, suggesting a deviation from pure thermionic emission, including the contribution from tunneling. While thermionic emission remains the dominant mechanism throughout the measured temperature range, the onset of thermionic field emission at low temperatures supplements the barrier inhomogeneity effects and is consistent with the markedly elevated ideality factor ($\eta \approx 7.41$) observed in region I. Current at low temperatures is preferentially channeled through lower-barrier patches, where the reduced thermal energy restricts sampling of the full barrier distribution.[28]

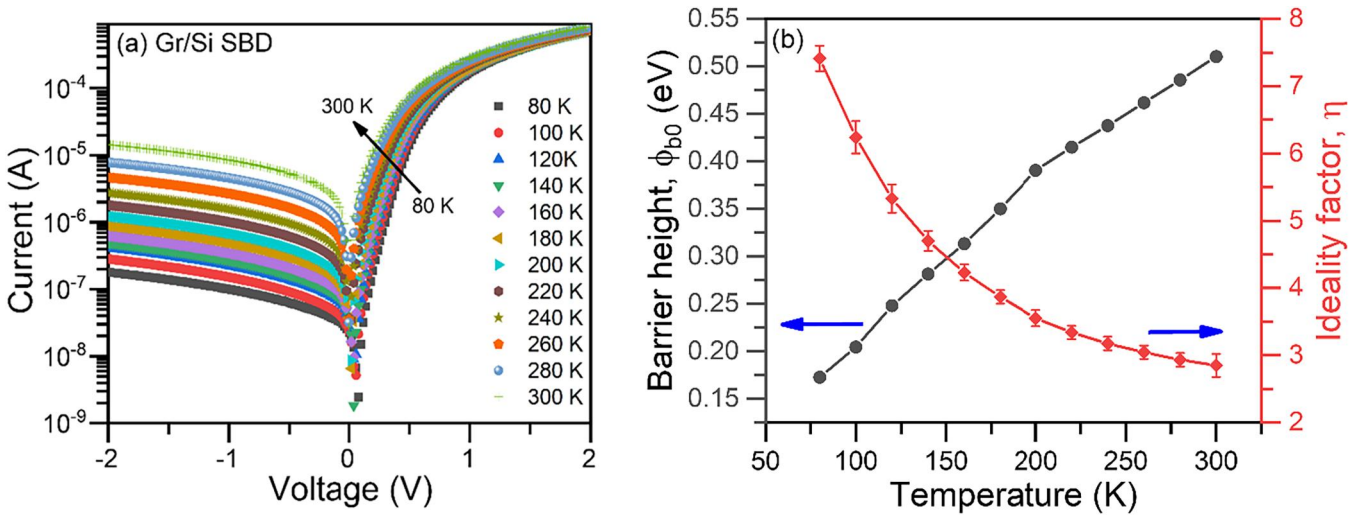


FIG. 4. (a) Temperature-dependent I–V characteristics of the Gr/Si Schottky barrier diode from 80 to 300 K. Curves shift systematically with temperature, consistent with thermally activated transport. (b) Temperature dependence of the ideality factor η and zero-bias Schottky barrier height ϕ_{b0}, showing a monotonic decrease in η and an increase in ϕ_{b0} with rising temperature, indicative of Schottky barrier inhomogeneity at the graphene/silicon interface. The error bars represent the statistical uncertainties obtained from the linear regression at each temperature.

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The series resistance R_s was extracted as a function of temperature by applying the Cheung and Cheung method[29] to the high-bias forward I–V data. Figure 5 shows the variation in R_s as a function of temperature. The series resistance remains nearly constant at approximately 1.70 kΩ over the temperature range of 80–260 K, with only minor fluctuations within the experimental uncertainty. At higher temperatures, a gradual reduction in R_s is observed, reaching approximately 1.61 kΩ at 300 K. The weak temperature dependence of R_s over most of the investigated range suggests that the resistive contribution from the non-Schottky regions of the device remains relatively stable, while the reduction at elevated temperatures may be associated with enhanced thermally activated carrier transport and reduced carrier freeze-out.[30]

The barrier height inhomogeneity was quantitatively analyzed using the Gaussian distribution formalism proposed by Werner and Güttler.[31] Under the assumption that ϕ_{b0} follows a Gaussian distribution with mean ϕ_m and standard deviation σ_s, the temperature dependence of the apparent barrier height is given by[20,31]

$$\phi_{b0} = \phi_m - \frac{q\sigma_s^2}{2k_BT}. \tag{4}$$

A plot of ϕ_{b0} vs $q/2k_BT$ yields a straight line with intercept ϕ_m and slope σ_s^2. The standard deviation is recovered directly from the slope of the linear fit. Similarly, the ideality factor varies with temperature as[31]

$$\frac{1}{\eta} - 1 = \rho_2 - \frac{q\rho_3}{2k_BT}, \tag{5}$$

where ρ_2 and ρ_3 are temperature-independent voltage coefficients characterizing bias-induced barrier deformation. Figure 6 shows the combined Werner–Güttler plot of ϕ_{b0} and $(\eta^{-1} - 1)$ vs $q/2k_BT$. Two distinct linear regions are clearly resolved: region I spanning 80–140 K and region II spanning 160–300 K, indicating that the barrier height distribution is best described by two overlapping Gaussian populations rather than a single distribution.

This dual-Gaussian behavior reflects the coexistence of two physically distinct populations of barrier-height fluctuations at the graphene/silicon interface. Region I, dominant at low temperatures, is attributed to transport through deep, isolated low-barrier patches associated with grain boundaries and structural defects in polycrystalline CVD graphene,[32] where the nascent thermionic field emission (TFE) contribution identified through the E_{00} analysis further supplements the current. Region II, dominant above ~160 K, reflects thermionic emission across the broader, more uniformly distributed barrier landscape governed by the mean graphene work function and interface dipoles introduced by the native SiO_2 layer.[17,33] The transition near 160 K is, therefore, not a sharp phase boundary but rather a temperature at which the thermal energy becomes sufficient to sample the full barrier distribution, progressively suppressing the contribution from deep low-barrier sites.[34]

From the linear fits in Fig. 6, the standard deviation of the Gaussian barrier distribution is determined to be $\sigma_s = 0.10$ eV in region I (80–140 K) and $\sigma_s = 0.05$ eV in region II (160–300 K). The larger σ_s in region I confirms stronger barrier inhomogeneity at low temperatures, consistent with the elevated ideality factor and the E_{00} analysis presented above.

To further quantify the role of interface states in the non-ideal transport behavior, the interface trap density D_{it} near the

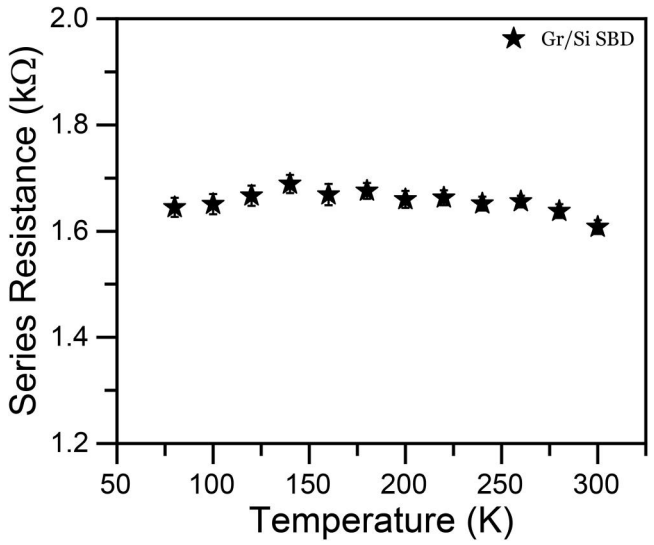


FIG. 5. Series resistance R_s vs temperature extracted from 80 to 300 K using the Cheung and Cheung method.[29] The series resistance remains nearly constant over most of the investigated temperature range, with a slight decrease observed at higher temperatures.

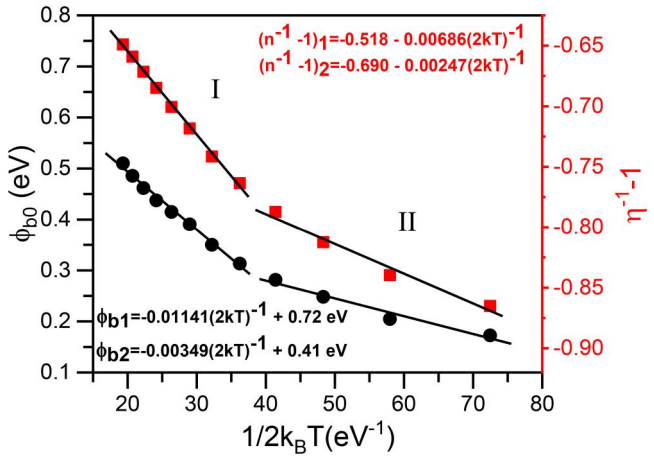


FIG. 6. Werner–Guttler plot showing the variation in zero-bias Schottky barrier height ϕ_{b0} (left axis) and $(\eta^{-1} - 1)$ (right axis) with $q/2k_BT$ for the Gr/Si Schottky barrier diode. Solid lines are linear fits to the data. Two distinct linear regimes are identified: region I (80–140 K) and region II (160–300 K), indicating a dual-Gaussian distribution of barrier heights at the graphene/silicon interface.

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Fermi level was estimated from the excess ideality factor using the capacitive approximation,[35,36]

$$D_{it} \approx \frac{(\eta - 1)\varepsilon_i}{q\delta_i}, \tag{6}$$

where $\epsilon_i \approx 3.9\ \epsilon_0$ is the permittivity of the interfacial SiO_2 layer, and the interfacial thickness is assumed to be $\delta_i \approx 1.5$ nm, consistent with values reported for intentionally regrown SiO_2 layers on Si.[37] At 300 K ($\eta = 2.84$), $D_{it} \approx 2.6 \times 10^{13}\ eV^{-1}\ cm^{-2}$, and at 80 K ($\eta = 7.41$), $D_{it} \approx 9.0 \times 10^{13}\ eV^{-1}\ cm^{-2}$. These values are consistent with amphoteric Pb^0-type defects at the Si/SiO_2 interface, which trap electrons under reverse bias and emit holes under forward bias.[8] The increase in D_{it} at lower temperatures reflects the progressive freezing out of thermally activated trap emission processes, causing a larger fraction of traps to remain charged and contribute to ideality factor enhancement at low temperatures.

The modified Richardson plot was constructed to extract the effective Richardson coefficient A* while accounting for barrier inhomogeneity.[21] By incorporating Eqs. (4) and (5) into Eq. (1), the modified expression is

$$\ln\left(\frac{I_0}{T^2}\right) - \frac{q^2\sigma_s^2}{2k_B^2T^2} = \ln(AA^*) - \frac{q\phi_m}{k_BT}. \tag{7}$$

Figure 7 shows the modified Richardson plot. Two linear segments corresponding to region I and region II are again clearly identified, consistent with the dual-Gaussian behavior established from Fig. 6. The slopes yield mean barrier heights $\phi_m = 0.42$ eV (region I) and 0.72 eV (region II), and the intercepts give $A^* = 21.2$ and $13.8\ A\ cm^{-2}\ K^{-2}$, respectively. Both values are substantially lower than the theoretical Richardson constant for n-type Si ($112\ A\ cm^{-2}\ K^{-2}$). This reduction is a direct consequence of the finite and energy-dependent density of states in graphene near the Dirac point, which limits the number of electronic states available for thermionic injection and renders the standard bulk Richardson formalism inapplicable for graphene/semiconductor contacts.[10] The Landauer transmission framework provides a more rigorous physical description of this effect, as discussed in detail by Sinha and Lee.[10]

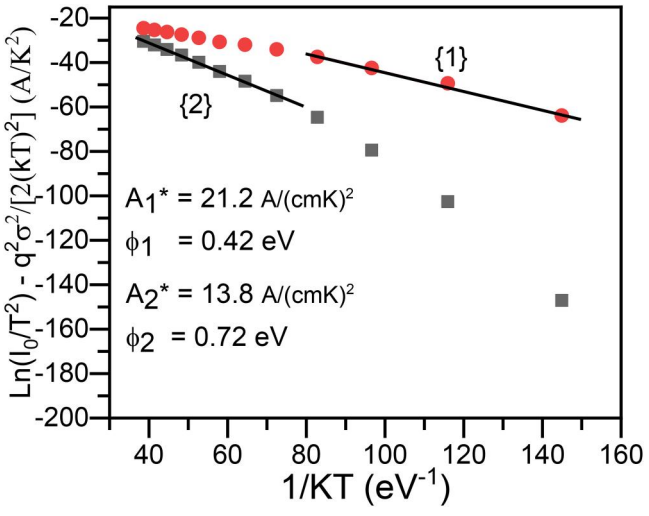


FIG. 7. Modified Richardson plot $[\ln(I_0/T^2) - q^2\sigma_s^2/2k_B^2T^{2z}]$ vs 1/T for the Gr/Si Schottky barrier diode. Two distinct linear regimes (region I: 80–140 K; region II: 160–300 K) yield mean barrier heights $\phi_m = 0.42$ and 0.72 eV, and Richardson constants $A^* = 21.2$ and $13.8\ A\ cm^{-2}\ K^{-2}$, respectively. Both values are significantly below the theoretical bulk value of $112\ A\ cm^{-2}\ K^{-2}$ for n-type Si, attributed to the finite density of states of graphene near the Dirac point.

CONCLUSIONS

The structural, optical, and electronic characterization confirmed the successful fabrication of high-quality CVD graphene for Schottky device applications. Raman spectroscopy verified the formation of predominantly monolayer graphene through the characteristic G and 2D bands, a high I_{2D}/I_G ratio of ~3.6, a narrow symmetric 2D peak, and a weak D band indicative of low defect density. UPS yielded a graphene work function of 5.02 eV, confirming the p-type nature of the transferred graphene and demonstrating its suitability as a transparent Schottky contact on n-type Si.

The temperature-dependent I–V characteristics of CVD graphene/n-type silicon Schottky barrier diodes were systematically investigated over the range of 80–300 K. At room temperature, the ideality factor and Schottky barrier height were determined to be $\eta = 2.84$ and $\phi_{b0} = 0.51$ eV, from fits to the thermionic emission model. The measured SBH falls within the range of 0.35–0.75 eV expected from the Schottky–Mott relation for graphene/n-Si contacts, confirming the formation of a well-defined rectifying junction. The room-temperature rectification ratio of approximately 10^2 at $|V| = 1$ V is consistent with previously reported values for CVD graphene/Si diodes. The ideality factor decreases monotonically from 7.41 at 80 K to 2.84 at 300 K, while the apparent SBH increases from 0.17 to 0.51 eV over the same range. This inverse temperature dependence is a definitive signature of Schottky barrier inhomogeneity at the graphene/silicon interface. The reverse leakage current increases from $\sim 10^{-7}$ A at 80 K to $\sim 10^{-5}$ A at 300 K, consistent with thermally activated carrier generation across an inhomogeneous barrier landscape. The series resistance, extracted using the Cheung and Cheung method, exhibits weakly non-monotonic temperature dependence, remaining nearly constant over most of the investigated temperature range with a slight reduction at higher temperatures. This behavior reflects the combined temperature-dependent contributions of the Si substrate, graphene, and graphene/Si interface.

Quantitative analysis within the Werner–Güttler Gaussian barrier distribution formalism reveals two distinct transport regimes separated by a transition near 160 K. The low-temperature regime (region I: 80–140 K) is attributed to tunneling-assisted transport through deep, localized low-barrier patches associated with grain boundaries and structural defects in polycrystalline CVD graphene, while the high-temperature regime (region II: 160–300 K) reflects thermionic emission across the broader, more uniform barrier landscape governed by the mean graphene work function and interface dipoles introduced by the native SiO_2 layer. The characteristic tunneling energy calculations revealed that

25 October 2026 07:43:04

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thermionic emission dominates throughout the measured range, with a nascent TFE contribution at the lowest temperatures that supplements barrier inhomogeneity effects in producing the elevated ideality factor in region I.

From the linear fits of the Werner–Güttler plots, the mean barrier heights are $\phi_m = 0.42$ eV (region I) and 0.72 eV (region II). The modified Richardson plot yields Richardson constants $A^* = 21.2\ A\ cm^{-2}\ K^{-2}$ (region I) and $13.8\ A\ cm^{-2}\ K^{-2}$ (region II), both substantially below the theoretical bulk Si value of $112\ A\ cm^{-2}\ K^{-2}$. This systematic reduction is a direct consequence of the finite, energy-dependent density of states near the Dirac point of graphene, which limits the available states for thermionic injection beyond what the bulk Richardson formalism assumes. The interface trap density lies in the range of ~2.6–$9.0 \times 10^{13}\ eV^{-1}\ cm^{-2}$ between 300 and 80 K, in agreement with the density of amphoteric Pb^0-type defects known to be present at the Si/SiO_2 interface. All key electrical parameters were reproducible within ~5% across three independently fabricated devices, confirming the robustness of the fabrication process and the generality of the reported transport behavior.

Taken together, these results establish a consistent and physically complete picture of charge transport in CVD graphene/n-Si Schottky barrier diodes, in which a dual-Gaussian barrier distribution, a finite graphene density of states, and thermally activated interface-trap processes collectively govern device behavior across the full 80–300 K range. The thermal stability of the rectifying characteristics, the manageable degree of barrier non-uniformity, and the well-defined transition between transport regimes highlight the suitability of graphene/Si Schottky diodes for deployment in temperature sensing, radiation detection, and moderate-power electronic applications where reliable operation over a wide thermal range is required. Realizing the diodes' potential for higher-power operation will, however, require careful thermal management: sustained current densities above ~100 mA in the graphene layer could otherwise cause localized heat-up and hotspot formation, so power-handling claims beyond the moderate-current regime studied here should be regarded as provisional pending dedicated high-current measurements.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

M.V. and B.C. contributed equally in this work.

Muskan Verma: Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Validation (equal); Visualization (equal); Writing – original draft (equal). **Bhumika Chaudhary:** Data curation (equal); Methodology (equal); Writing – original draft (equal). **Rakesh Kumar:** Conceptualization (equal); Data curation (equal); Supervision (equal); Writing – review & editing (equal). **Ashish Kumar:** Conceptualization (equal); Data curation (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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